

Home Work 1
II B. Tech. (Common)
Mathematics-II (NMCI102)

1. Solve the following systems of linear equations (if possible).

(a) $x_1 + x_2 = 4$, $x_2 - x_3 = 1$, $2x_1 + x_2 + 4x_3 = 7$ (b) $x_1 + 3x_2 + x_3 = 0$, $2x_1 - x_2 + x_3 = 0$

(c) $x_1 + 2x_2 - x_3 = 10$, $-x_1 + x_2 + 2x_3 = 2$, $2x_1 + x_2 - 3x_3 = 2$

(d) $x + y + z - 3w = 1$, $2x + 4y + 3z + w = 3$, $3x + 6y + 4z - 2w = 4$

(e) $x_1 + 2x_2 - x_3 = 10$, $-x_1 + x_2 + 2x_3 = 2$, $2x_1 + x_2 - 3x_3 = 8$.

(f) $x + y + z + 2w = 1$, $x + 2y + z + w = 2$.

Answers: (a) (3,1,0) (b) $c(-4/7, -1/7, 1)$, $c \in \mathbb{R}$, (c) inconsistent

(d) $(0, 0, 1, 0) + d(10, 0, -7, 1)$, $d \in \mathbb{R}$ (e) $(2, 4, 0) + c(5/3, -1/3, 1)$, $c \in \mathbb{R}$ (f) $(0, 1, 0, 0) + c(-1, 0, 1, 0) + d(-3, 1, 0, 1)$, $c \in \mathbb{R}$.

2. Determine the conditions for which the following system $x + y + z = 1$, $x + 2y - z = b$, $5x + 7y + az = b^2$ admits (a) only one solution (b) no solution (c) many solutions.

Answers: (a) $a \neq 1$ (b) $a = 1, b \neq -1, 3$ (c) $a = 1, b = -1$ or $b = 3$.

3. Consider the system:

$$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu,$$

where $\lambda, \mu \in \mathbb{R}$. Determine the conditions on λ and μ for which the system admits (a) only one solution (b) no solution (c) many solutions.

4. Using the row echelon form of a matrix, find the rank of the following matrices. (a) $A = \begin{bmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{bmatrix}$ (b)

$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 0 \end{bmatrix}$ (c) $A = \begin{bmatrix} -2 & 0 & 0 & 3 \\ 1 & 5 & 3 & 0 \\ 3 & 2 & 1 & 6 \\ 3 & 5 & 3 & -3 \end{bmatrix}$.

Answers: (a) 2 (b) 2 (c) 3

5. Find the rank of the matrix $A = \begin{bmatrix} a & -1 & 1 \\ -1 & a & -1 \\ -1 & -1 & a \\ 1 & 1 & 1 \end{bmatrix}$ if (a) $a \neq -1$ (b) $a = -1$. **Answers:** (a) 3 (b) 2.

6. Find x such that the rank of the matrix $A = \begin{bmatrix} 1 & 3 & -3 & x \\ 2 & 2 & x & -4 \\ 1 & 1-x & 2x+1 & -5-3x \end{bmatrix}$ is 2. **Answers:** There does not exist any value of $x \in \mathbb{R}$.

7. Find the value of a for which the rank of the matrix $\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & a & 1 & 0 \\ -1 & 1 & 1 & -1 \end{bmatrix}$ is 3.

8. Consider the system: $3x - y + z = 1$, $2x + y - z = 2$, $x - 2y + 2z = 5$. Find the rank of the coefficient matrix and the augmented matrix. Hence, determine the consistency of this system.

9. Let P be a non-zero matrix of order $m \times 1$ and Q be a non-zero matrix of order $1 \times n$. Determine $\text{rank}(PQ)$.

10. Let P be a matrix of order $m \times n$ and Q be a matrix of order $n \times p$, $n \neq p$. If $\text{rank}(P) = n$ and $\text{rank}(Q) = p$, determine $\text{rank}(PQ)$.

11. Using the row reduced echelon form of a matrix (Gauss-Jordan Method), find the inverse of the following matrices.

(a) $\begin{bmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{bmatrix}$.

Answers: (a) $\begin{bmatrix} 1/8 & -5/8 & 3/4 \\ -1/4 & 3/4 & -1/2 \\ 38 & -3/8 & 1/4 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & -1/2 & -2 \\ -1 & 1/2 & 0 \\ -3 & 0 & 1 \end{bmatrix}$.